

Calculus Review

Log Rules

Rule	Example
1) $\ln(xy) = \ln(x) + \ln(y)$	$\ln(5 \times 3) = \ln(5) + \ln(3)$
2) $\ln\left(\frac{x}{y}\right) = \ln(x) - \ln(y)$	$\ln\left(\frac{5}{3}\right) = \ln(5) - \ln(3)$
3) $A \ln(x) = \ln(x^A)$	$5 \ln(3) = \ln(3^5) = \ln(243)$

Exponent Rules

Rule	Example
1) $x^a x^b = x^{a+b}$	$x^5 x^3 = x^8$
2) $(x^a)^b = x^{ab}$	$(x^5)^3 = x^{15}$
3) $\frac{x^a}{x^b} = x^{a-b}$	$\frac{x^5}{x^3} = x^2$

Derivative Rules

Type	Function	Derivative
Power Function	$y = Ax^n$	$\frac{dy}{dx} = nAx^{n-1}$
<i>Example</i>	$y = 5x^3$	$\frac{dy}{dx} = 15x^2$
Log Function	$y = A \ln(x)$	$\frac{dy}{dx} = \frac{A}{x}$
<i>Example</i>	$y = 7 \ln(x)$	$\frac{dy}{dx} = \frac{7}{x}$
Product Rule	$y = f(x)g(x)$	$\frac{dy}{dx} = f(x)g'(x) + g(x)f'(x)$
<i>Example</i>	$y = \sqrt{x}(x^3 + 4)$ Note: $f(x) = \sqrt{x}$, $g(x) = (x^3 + 4)$	$\frac{dy}{dx} = \sqrt{x}(3x^2) + (x^3 + 4)\frac{1}{2}x^{-1/2}$
Quotient Rule	$y = \frac{f(x)}{g(x)}$	$\frac{dy}{dx} = \frac{g(x)f'(x) - f(x)g'(x)}{g(x)^2}$
<i>Example</i>	$y = \frac{\sqrt{x}}{(x^3 + 4)}$	$\frac{dy}{dx} = \frac{(x^3 + 4)\frac{1}{2}x^{-1/2} - \sqrt{x}(3x^2)}{(x^3 + 4)^2}$
Chain Rule:	$y = f(g(x))$	$\frac{dy}{dx} = \frac{df}{dg} \frac{dg}{dx} = f'(g)g'(x)$
<i>Example</i>	$y = \ln(2x^7)$ Note: $f(g) = \ln(g)$, $g(x) = 2x^7$	$\frac{dy}{dx} = \frac{14x^6}{2x^7}$

Exponent Rule:	$y = a^x$	$\frac{dy}{dx} = a^x \ln a$
	$y = a^{f(x)}$	$\frac{dy}{dx} = a^{f(x)} f'(x) \ln a$
Special Case	$y = e^{f(x)}$	$\frac{dy}{dx} = e^{f(x)} f'(x) \ln e = e^{f(x)} f'(x)$
Example 1:	$y = 5^{x^3}$	$\frac{dy}{dx} = 5^{x^3} 3x^2 \ln 5$
Example 2:	$y = e^{(x^2+3x+5)}$	$\frac{dy}{dx} = e^{(x^2+3x+5)} (2x+3)$

Partial Derivatives:	$y = f(x_1, x_2, x_3, \dots, x_k)$	$\frac{\partial y}{\partial x_1} = \left. \frac{\partial f}{\partial x_1} \right _{x_2, \dots, x_n} = f_1$
Example 1:	$y = 3x_1^5 + x_1^2 - 4x_2^2 - 2x_1x_2$	$\frac{\partial y}{\partial x_1} = f_1 = 15x_1^4 + 2x_1 - 2x_2$
Example 2:	$y = e^{(7x_1^3x_2^2+3x_1+5x_2)}$	$\frac{\partial y}{\partial x_2} = f_2 = e^{(7x_1^3x_2^2+3x_1+5x_2)} (14x_1^3x_2 + 5)$

2nd Order Partial Derivatives:

	$y = f(x_1, x_2, x_3, \dots, x_k)$	$\frac{\partial^2 y}{\partial x_i \partial x_j} = f_{ij}$
Example 1:	$y = 3x_1^5 + x_1^2 - 4x_2^2 - 2x_1x_2$	$f_{11} = 60x_1^3 + 2$ $f_{12} = -2$
Example 2:	$y = e^{(7x_1^3x_2^2+3x_1+5x_2)}$	$f_{22} = e^{(7x_1^3x_2^2+3x_1+5x_2)} 14x_1^3 + (14x_1^3x_2 + 5)e^{(7x_1^3x_2^2+3x_1+5x_2)} (14x_1^3x_2 + 5)$ $f_{21} = e^{(7x_1^3x_2^2+3x_1+5x_2)} 42x_1^2x_2 + (14x_1^3x_2 + 5)e^{(7x_1^3x_2^2+3x_1+5x_2)} (21x_1^2x_2 + 3)$

Young's Theorem

$$\frac{\partial^2 y}{\partial x_i \partial x_j} = \frac{\partial^2 y}{\partial x_j \partial x_i} \text{ or } f_{ij} = f_{ji}$$

Total Differential

Univariate Case

Function: $y = f(x)$

Total Differential: $dy = \frac{\partial f}{\partial x} dx = f_x dx$

Multivariate Case

Function: $y = f(x_1, x_2, x_3, \dots, x_k)$

Total Differential: $dy = f_1 dx_1 + f_2 dx_2 + \dots + f_n dx_n$

Implicit Functions and Their Derivatives

Given $F(y, x_1, x_2, x_3, \dots, x_k) = 0$, the total differential of F is:

$$F_y dy + F_1 dx_1 + F_2 dx_2 + \dots + F_n dx_n = 0$$

In order to calculate the derivative $\frac{dy}{dx_1}$, holding the other variables constant, one need merely manipulate the total differential to show:

$$\frac{dy}{dx_1} = -\frac{F_{x_1}}{F_y}$$

Example

Suppose you have a production possibility frontier of the form $5x^2 + 3y^2 = 768$.

This equation could be written implicitly as $F(x, y) = 5x^2 + 3y^2 - 768 = 0$.

To derive the slope of the production possibility frontier one can use the implicit function theorem as follows:

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{10x}{6y} = -\frac{5x}{3y}$$

Interpretation: At the point (9,11) the slope is $-45/33$, or approximately -1.36. This means that if x decreased 1 unit from that point, y must increase by 1.36 in order to stay on the PPF.

Optimization Rules

Univariate Optimization

Objective Function

$$y = f(x)$$

First Order Condition

$$\frac{dy}{dx} = f'(x) = 0$$

F.O.C.

Necessary Condition

Second Order Condition

$$\frac{d^2y}{dx^2} = f''(x) < 0 \quad \text{Maximum}$$

S.O.C.

Sufficient Condition

$$\frac{d^2y}{dx^2} = f''(x) > 0 \quad \text{Minimum}$$

Example

Objective Function

$$y = x^4 + 108x - 5$$

First Order Condition

$$\frac{dy}{dx} = 4x^3 + 108 = 0, \text{ satisfied when } x = -3$$

Second Order Condition

$$\frac{d^2y}{dx^2} = 12x^2 = 12(-3)^2 = 108 > 0 \quad \therefore \text{minimum}$$

Optimization Rules

Multivariate Optimization

Objective Function

$$y = f(x_1, x_2, x_3, \dots, x_k)$$

First Order Conditions

$$\frac{\partial y}{\partial x_1} = \frac{\partial y}{\partial x_2} = \frac{\partial y}{\partial x_3} = \dots = \frac{\partial y}{\partial x_k} = 0$$

Second Order Conditions

Depends upon the sign (+/-) of the determinant of the Hessian matrix
See Chapter 11 of A. Chiang, Fundamental Methods of Mathematical Economics, 3rd edition.

(Two Variable Case)

$$\text{Maximum: } f_{xx}, f_{yy} < 0 \text{ and } f_{xx}f_{yy} > f_{xy}^2$$

$$\text{Minimum: } f_{xx}, f_{yy} > 0 \text{ and } f_{xx}f_{yy} > f_{xy}^2$$

Example

Objective function:

$$f(x, y) = 60 + 8x + 2y - x^2 - 0.5y^2$$

First order conditions:

$$f_x = 8 - 2x = 0, \quad f_y = 2 - y = 0, \quad \Rightarrow \quad x^* = 4, y^* = 2$$

Second order conditions:

$$f_{xx} = -2, \quad f_{yy} = -1, \quad f_{xy} = 0$$

Evaluated at the optimal values:

$$f_{xx} = -2, \quad f_{yy} = -1, \quad f_{xy} = 0$$

Hence,

$$f_{xx} < 0, \quad f_{yy} < 0, \quad f_{xx}f_{yy} > f_{xy}^2 \quad \Rightarrow \quad \text{a maximum}$$

The Envelope Theorem

The envelope theorem is concerned with how the optimal value of a function changes when a parameter of the function changes. The theorem says that finding how the optimal value changes with a parameter change is as easy as finding how the function changes with a parameter change (when you evaluate that change at the optimal value).

For example, suppose the objective function is:

$$y = f(x, \alpha)$$

where x = variable
 α = parameter

If we optimize $f(x, \alpha)$ by choosing x we will have the optimal value of x being a function of α -- this happens through solving the first order condition:

$$\text{F.O.C. } \frac{dy}{dx} = \frac{df}{dx} = 0 \Rightarrow x^* = g(\alpha)$$

Substituting x^* into the objective function gives us the indirect objective function:

$$y^* = f(x^*(\alpha), \alpha)$$

If we want to know how y^* changes as α changes, we would need to take the following derivative:

$$\frac{\partial y^*}{\partial \alpha} = \frac{\partial f}{\partial x} \frac{\partial x}{\partial \alpha} + \frac{\partial f}{\partial \alpha}$$

Note that at the optimal value of y^* , $\frac{\partial f}{\partial x} = 0$ by the first order conditions. Hence,

when $\frac{\partial y^*}{\partial \alpha}$ is evaluated at x^* the above equation collapses to:

$$\frac{\partial y^*}{\partial \alpha} = \frac{\partial f(x, \alpha)}{\partial \alpha}$$

The next reasonable question is, "What does this have to do with an envelope?" The answer to that has to do with slopes and tangent lines. As you know, the derivative is the slope or tangent to a function. If you mapped the function $y^* = f(x^*, \alpha)$ and then varied α , you would find that $\frac{\partial y^*}{\partial \alpha}$ would provide the tangent lines to the function y^* making an effective "envelope" of the function. See the diagram on p. 39 of the Nicholson text.

An Application of the Envelope Theorem

Profit Maximization:

p = price of product

q = quantity of product = $x_1 x_2$

x_i = input into production

w_i = price of input i

Profit Function : $\pi(x_1, x_2, p, w_1, w_2) = TR - TC = px_1 x_2 - x_1 w_1 - x_2 w_2$

In this problem we maximize profits by choosing input levels, (x_1, x_2) , given the cost of the inputs (w_1, w_2) . The input costs are parameters in the above equation.

Method 1: If you want to know how the optimal value of profits changes as the input price of good 1 changes, you could invoke the envelope theorem to calculate:

$$\frac{\partial \pi^*}{\partial w_1} = -x_1^*$$

Method 2: Or you could go through the entire optimization problem as follows:

$$\begin{array}{ll} \frac{\partial \pi}{\partial x_1} = px_2 - w_1 = 0 & x_2^* = \frac{w_1}{p} \\ \text{F.O.C.} & \Rightarrow \\ \frac{\partial \pi}{\partial x_2} = px_1 - w_2 = 0 & x_1^* = \frac{w_2}{p} \end{array}$$

It is clear that $x^* = x^*(w, p)$ such that the optimal profit function is written as follows:

$$\pi^*(x_1^*(w, p), x_2^*(w, p), w_1, w_2, p) = px_1^* x_2^* - x_1^* w_1 - x_2^* w_2$$

and the derivative of π^* with respect to w_1 would be written as:

$$\begin{aligned} \frac{\partial \pi^*}{\partial w_1} &= (px_1^*) \frac{\partial x_2^*}{\partial w_1} + px_2^* \frac{\partial x_1^*}{\partial w_1} - x_1^* - w_1 \frac{\partial x_1^*}{\partial w_1} - w_2 \frac{\partial x_2^*}{\partial w_1} \\ &= \frac{px_1^*}{p} + 0 - x_1^* - 0 - \frac{w_2}{p} = -x_1^* \end{aligned}$$

Note that methods 1 and 2 provide the same solution but 1 was a lot easier.